generated_report_1
pbmc_multiome: objective-aware integration

> **Integration verdict: integrate_with_caution**
>
> The validation-selected representation is clip_dual (clip_cross_modal). Its selected encoders are RNA MLP head (PCA input), ATAC MLP head (TF-IDF/LSI input), with projected_mean fusion and symmetric_InfoNCE objective. Held-out comparisons and uncertainty appear below.

Objective and principal findings

Learn a joint RNA+ATAC cell representation that preserves PBMC cell-type structure at least as well as the better single modality, aligns the modalities so that a cell's RNA profile retrieves its own ATAC profile, and supports cross-modal prediction of chromatin accessibility from expression. Cell-type labels may be used for evaluation only, not for fitting the primary representation.

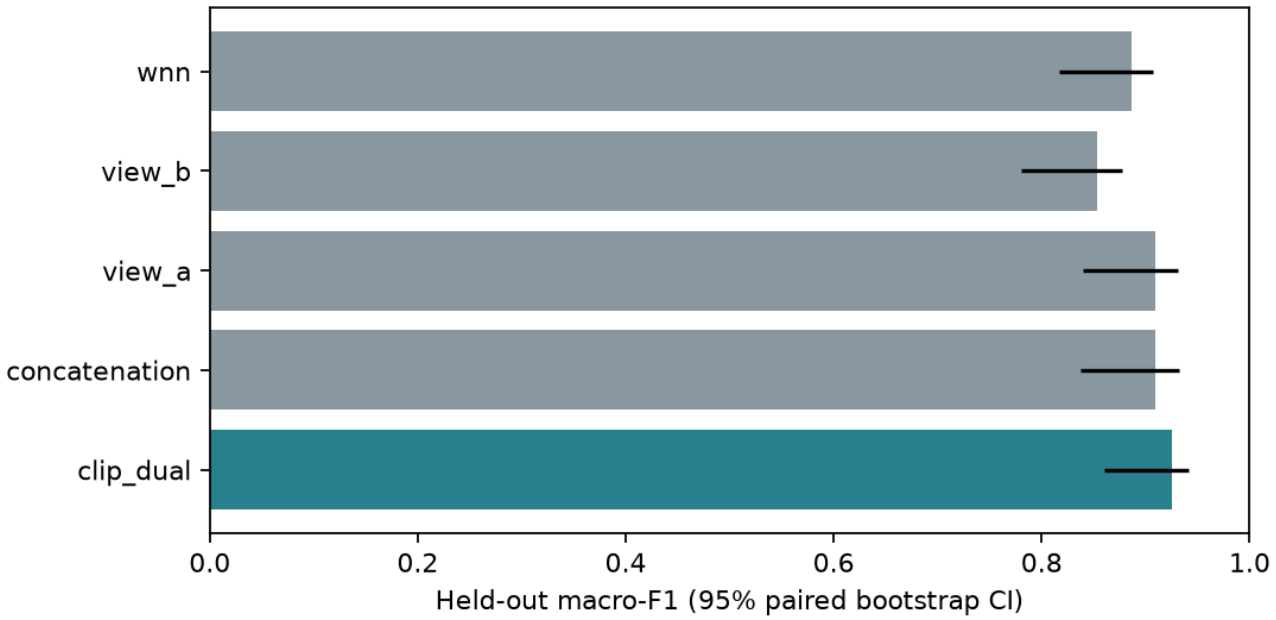
Allowed supervision: **evaluation_only**. Primary metrics: macro_f1, retrieval_recall_at_10. The deterministic planner and templated reporting used no external LLM.

Representation	Family	Dim.	Macro-F1	95% CI	Balanced accuracy
clip_dual	clip_cross_modal	32	0.925	0.861–0.941	0.917
concatenation	baseline_fusion	64	0.910	0.837–0.932	0.890
view_a	baseline_unimodal	32	0.909	0.840–0.931	0.887
view_b	baseline_unimodal	32	0.853	0.780–0.877	0.831
wnn	baseline_fusion	32	0.886	0.817–0.907	0.854

All rows use the same frozen-embedding logistic probe protocol. End-to-end head predictions are a separate analysis. The selected model was frozen before test access. Paired bootstrap resampling follows the sample unit recorded in the evidence.

Baseline names: view_a uses rna; view_b uses atac. Concatenation uses the scaled feature blocks from both modalities.

Against the best single modality (view_a), the selected-minus-reference macro-F1 difference is 0.016 (95% CI -0.009 to 0.060). Project decision thresholds are pre-registered in METHOD_NOTES.md and are not external scientific standards.



Held-out comparison; candidate selection used validation only.

Data, pairing and cohort

Modality	Type	Samples	Features	Missingness/QC
rna	count_matrix_gene	11909	36601	sparse counts; missing covariates tracked separately
atac	count_matrix_peak	11909	108377	sparse counts; missing covariates tracked separately

Pairing level: cell. All eligible samples were assigned before fitting. Single donor: no donor generalization. Curated FixFlip labels derive from WNN clusters; WNN structural advantage; evaluation-only labels.

Cohort step	Count
cells before agent qc	11909
curated labeled	10412

Preprocessing and leakage control

Fit scope: training only. Preprocessing receipts record fitted identifiers, split hashes, QC thresholds and feature selection. Neither test data nor target labels fit preprocessing.

Modality	Method	Retained features
rna	Seurat HVGs on training log-normalized RNA; scaled/clipped PCA	3000
atac	train-frequency peak filter; binary TF-IDF (log1p 1e4 TF*IDF); LSI	30000

Raw expression counts are retained for count likelihoods. First-LSI depth correlation: 0.932; dropped first component: True.

Agent QC cohort step	Cells
after	10243
before	11909

Agent QC cohort step	Cells
labeled passing	10078
vignette filtered passing	165
vignette labeled rejected	334
test after QC	1540
train after QC	7171
validation after QC	1532

QC variable	Scale	Lower bound	Upper bound
atac fragments	log1p	7.925	11.085
fraction fragments in peaks	identity	0.601	Not set
mito fraction	identity	0.000	0.200
rna genes	log1p	6.289	8.728
rna umis	log1p	6.555	9.919
tss fragment fraction	identity	0.218	Not set

Doublet decision: Diagnostic only: no validated out-of-sample doublet scoring; QC exclusion uses count/QC thresholds. TSS measure: TSS fragment fraction proxy; not conventional flanking-normalized TSS enrichment.

Training doublet-score threshold 0.178; predicted training doublet fraction 0.099. This diagnostic does not exclude cells.

Agent choices and empirical pilot

Candidate	Status	Rule and reason
view_a	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
view_b	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
concatenation	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
wnn	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
multivi	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
mofa	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
simclr	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
clip_dual	eligible for pilot	ELIGIBLE_PROFILE_OBJECTIVE: Compatible modalities and objective
fusion_early	reference only	LABELS_EVALUATION_ONLY: Objective does not allow representation fitting on labels

Candidate	Status	Rule and reason
fusion_late	reference only	LABELS_EVALUATION_ONLY: Objective does not allow representation fitting on labels
fusion_gated	reference only	LABELS_EVALUATION_ONLY: Objective does not allow representation fitting on labels
abmil	not applicable	WSI_ENCODER_REQUIRED: No whole-slide image modality
fm_last_block	not applicable	WSI_ENCODER_REQUIRED: No whole-slide image modality
fm_image_text	not applicable	WSI_ENCODER_REQUIRED: No whole-slide image modality
histology_conch	not applicable	WSI_ENCODER_REQUIRED: No whole-slide image modality
histology_imagenet	not applicable	WSI_ENCODER_REQUIRED: No whole-slide image modality

Selection rule: EPSILON_PARETO_SEQUENTIAL_QUALITY_CONSTRAINTS_THEN_SIMPLICITY. Validation shortlist: clip_dual. Separate evidence columns preserve utility, alignment, structure, validity, stability and compute rather than an unexplained weighted score.

Pilot caps: 600 adapter-fit seconds and 50 epochs/iterations; declared seeds: 774, 775, 776. Every eligible candidate is attempted. Import and data-loading time are excluded from adapter fit time. Shared PCA/LSI fitting occurs in preprocessing; this is not a claim of zero total baseline cost.

Pilot candidate	Successful / attempted	Validation macro-F1 (seed SD)	Recall@10	Failure evidence
clip_dual	3 / 3	0.916 (0.004)	0.510	None
concatenation	1 / 1	0.925	0.439	None
mofa	3 / 3	0.887 (0.000)	0.395	None
multivi	3 / 3	0.716 (0.042)	0.133	None
simclr	3 / 3	0.925 (0.002)	0.342	None
view_a	1 / 1	0.902	0.439	None
view_b	1 / 1	0.808	0.439	None
wnn	1 / 1	0.904	0.263	None

Primary-metric differences after freezing

Metric	Selected minus best single (95% CI)	Selected minus simple fusion (95% CI)
macro_f1	0.016 (-0.009, 0.060)	0.015 (-0.007, 0.059)
retrieval_recall_at_10	0.069 (0.050, 0.090)	0.069 (0.050, 0.090)

Model	Seed	Fit seconds	Epochs/steps	Stopping
view_a	774	0.0	0	family-specific convergence/max iterations
view_b	774	0.0	0	family-specific convergence/max iterations
concatenation	774	0.0	0	family-specific convergence/max iterations
wnn	774	58.6	0	family-specific convergence/max iterations
clip_dual	774	8.7	59	validation early stopping
clip_dual	775	7.1	48	validation early stopping

Model	Seed	Fit seconds	Epochs/steps	Stopping
clip_dual	776	10.4	76	validation early stopping

Project aggregate GPU time in the saved ledger: 2.770 GPU-hours (reconciled 2026-11-03T11:00:33.473861+00:00). This includes both datasets and runtime/throughput acceptance work, rather than only this dataset. The immutable ledger snapshot accompanies this report.

Representation quality and confounding

Model	Effective rank	Leiden ARI	Leiden NMI	Alignment
clip_dual	22.66	0.528	0.777	FOSCTTM 0.016; pool 1540
concatenation	55.42	0.679	0.807	FOSCTTM 0.023; pool 1540
view_a	31.68	0.613	0.756	FOSCTTM 0.023; pool 1540
view_b	29.86	0.576	0.717	FOSCTTM 0.023; pool 1540
wnn	30.27	0.451	0.724	FOSCTTM 0.043; pool 1540

Graph connectivity, global-distance correlations, each modality's neighborhood preservation, per-class results, QC/depth or site predictability, seed/subsample stability and modality ablations are saved as distinct evidence categories.

Alignment definitions from the frozen run records: concatenation, view_a, view_b: train-fitted ridge cross-view alignment probe. clip_dual, wnn: adapter per-modality embeddings in shared latent coordinates. A ridge alignment probe uses training pairs to map the first embedding into the second embedding's coordinates; its retrieval scores describe that inductive cross-view probe. They do not establish that independently learned or single-modality embeddings intrinsically share a coordinate system.

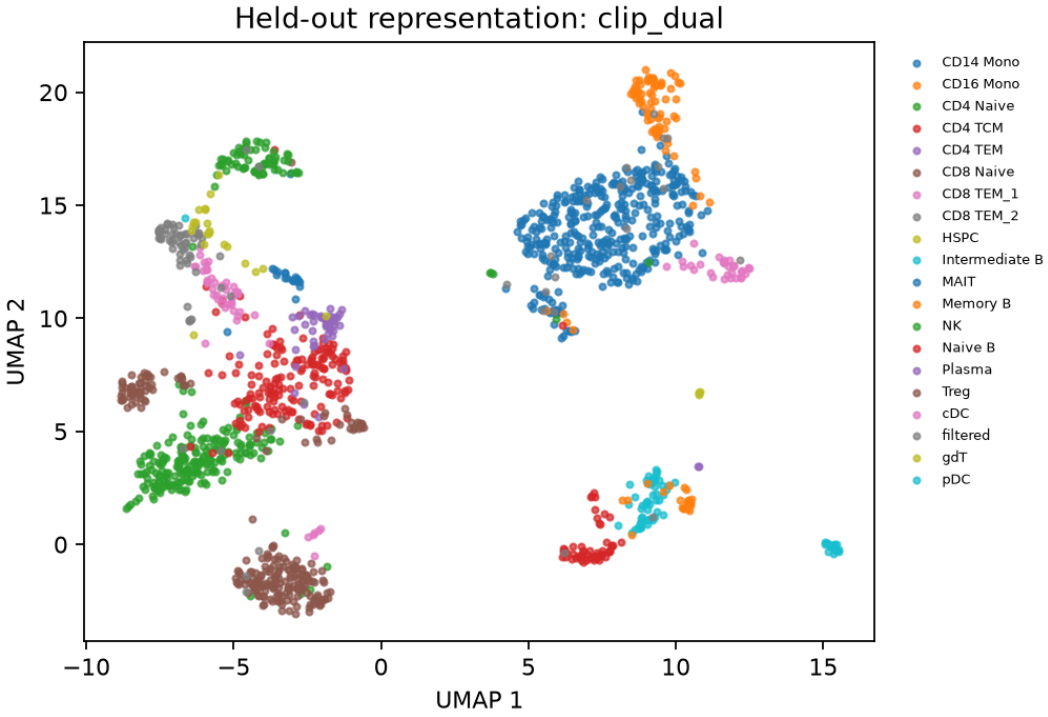
Selected-model class	Support	Recall (95% CI)	OVR AUROC (95% CI where evaluated)	Graph connectivity
CD14 Mono	380	0.987 (0.975, 0.995)	1.000	1.000
CD16 Mono	77	0.922 (0.870, 0.974)	0.999	1.000
CD4 Naive	212	0.929 (0.896, 0.962)	0.997	0.972
CD4 TCM	171	0.877 (0.825, 0.924)	0.992	0.982
CD4 TEM	45	0.889 (0.778, 0.978)	0.998	1.000
CD8 Naive	209	0.995 (0.986, 1.000)	0.999	1.000
CD8 TEM_1	48	0.896 (0.812, 0.979)	0.998	1.000
CD8 TEM_2	54	0.870 (0.778, 0.963)	0.998	0.981
HSPC	4	1.000 (1.000, 1.000)	1.000	1.000
Intermediate B	52	0.981 (0.942, 1.000)	1.000	1.000

Selected-model class	Support	Recall (95% CI)	OVR AUROC (95% CI where evaluated)	Graph connectivity
MAIT	21	0.905 (0.762, 1.000)	1.000	1.000
Memory B	21	0.857 (0.714, 1.000)	1.000	1.000
NK	70	0.986 (0.943, 1.000)	1.000	1.000
Naive B	55	1.000 (1.000, 1.000)	1.000	1.000
Plasma	2	1.000 (1.000, 1.000)	1.000	1.000
Treg	24	0.667 (0.500, 0.833)	0.997	1.000
cDC	29	0.897 (0.793, 1.000)	1.000	1.000
gdT	22	0.773 (0.591, 0.909)	0.997	0.909
pDC	16	1.000 (1.000, 1.000)	1.000	1.000

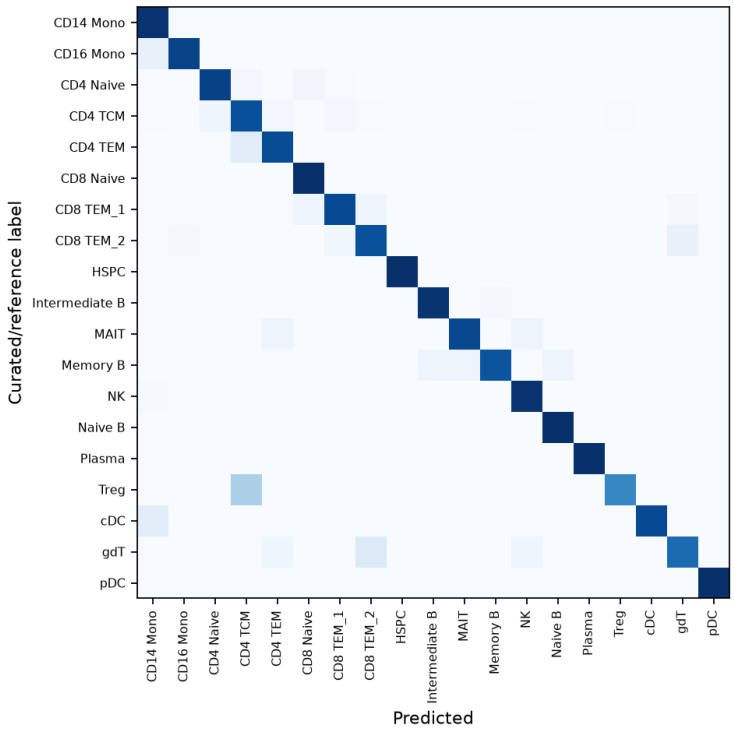
These intervals resample the observed test units. A rare class with only correct observed predictions can have a degenerate recall interval; this does not establish perfect population recall. Interpret class-level estimates together with their support.

Confounding diagnostic	Held-out score	Reference
atac_fragments	R^2 0.476	Train-fitted linear ridge predictor
mito_fraction	R^2 0.368	Train-fitted linear ridge predictor
rna_umis	R^2 0.703	Train-fitted linear ridge predictor
tss_fragment_fraction	R^2 0.864	Train-fitted linear ridge predictor

Validity/confounding cautions: Selected embedding predicts QC/depth with test R-squared >0.5; technical versus biological association unresolved



selected umap. UMAP is a train-fitted visualization and is not selection evidence.



confusion matrix. Held-out predictions from the frozen model; colors are row-normalized reference-class proportions.

Downstream analyses and sensitivity

Executed downstream tasks: classification, clustering, retrieval, cross_modal_prediction. Full machine-readable results accompany this report.

RNA→ATAC predictor	Per-cell AUROC	Per-cell AUPRC	Per-peak AUPRC
clip_dual_knn_decoder	0.801	0.506	0.286

RNA→ATAC predictor	Per-cell AUROC	Per-cell AUPRC	Per-peak AUPRC
concatenation_knn_decoder	0.800	0.506	0.284
knn_rna_pca	0.800	0.506	0.284
peak_frequency	0.779	0.511	0.211
view_a_knn_decoder	0.800	0.506	0.284
view_b_knn_decoder	0.800	0.506	0.284
wnn_knn_decoder	0.800	0.506	0.285

Accessibility targets are binarized peaks selected using train frequency 1%-99%, at most 1000 stratified by frequency. Held-out ATAC counts and features are masked before prediction. The separately named kNN decoders use each frozen model's RNA-side representation; the mean AUPRC across cells and the mean AUPRC across peaks summarize different rankings.

Sensitivity analyses: baseline_subsample_stability; coarse_labels; r_wnn_crosscheck; seed_stability; subsample_stability; . Read their separate metrics and fit scopes in downstream/results.json; unresolved analyses remain explicit.

Cross-modal alignment in the fixed held-out candidate pool

Model	A→B recall@1	A→B recall@5	A→B recall@10	B→A recall@10	Pool
clip_dual	0.127	0.356	0.500	0.494	1540
concatenation	0.090	0.252	0.380	0.476	1540
view_a	0.090	0.252	0.380	0.476	1540
view_b	0.090	0.252	0.380	0.476	1540
wnn	0.040	0.151	0.255	0.238	1540

Selected representation across the predeclared seeds

Seed	Validation macro-F1	Test macro-F1	Test balanced accuracy
774	0.916	0.925	0.917
775	0.911	0.928	0.924
776	0.918	0.919	0.911

Training-subsample stability, conditional on fixed preprocessing

Model	Seed	Training fraction	Validation neighbor overlap
clip_dual	774	0.800	0.837
clip_dual	775	0.800	0.775
clip_dual	776	0.800	0.790
concatenation	774	0.800	0.991
concatenation	775	0.800	0.991
concatenation	776	0.800	0.991
view_a	774	0.800	0.989

Model	Seed	Training fraction	Validation neighbor overlap
view_a	775	0.800	0.989
view_a	776	0.800	0.990
view_b	774	0.800	0.993
view_b	775	0.800	0.994
view_b	776	0.800	0.994
wnn	774	0.800	0.750
wnn	775	0.800	0.737
wnn	776	0.800	0.739

Refitted sensitivity analyses

Analysis	Macro-F1	Balanced accuracy
coarse labels	0.984	0.981

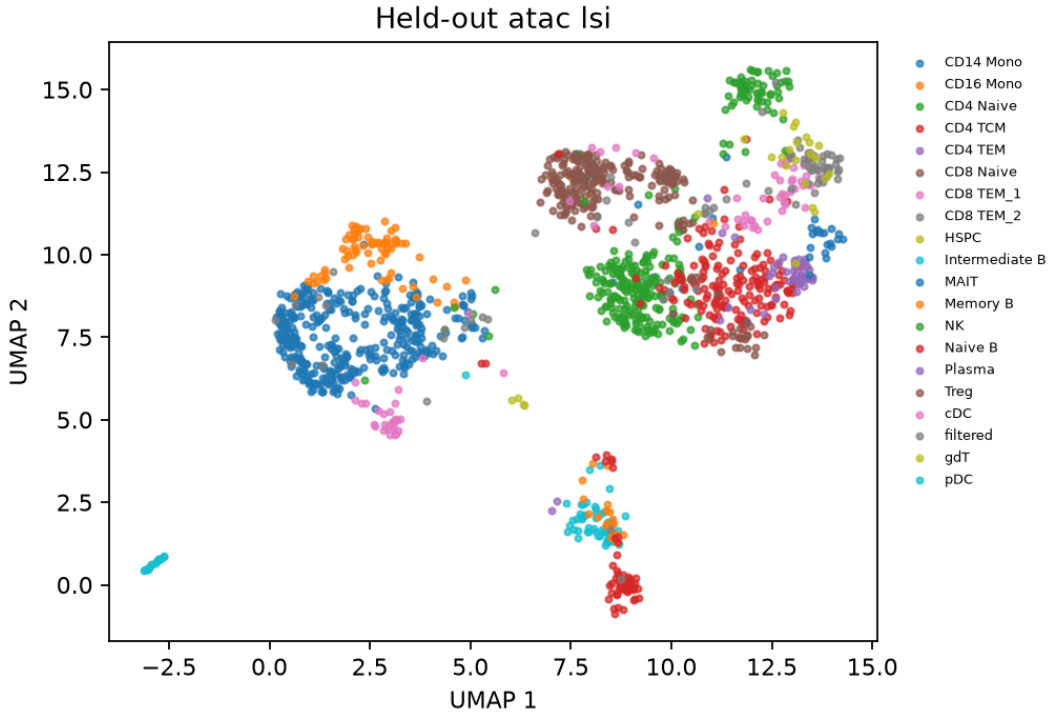
Patient/cell neighbors in the joint representation

Candidate pool	Label agreement@5	Label agreement@10
1540	0.894	0.883

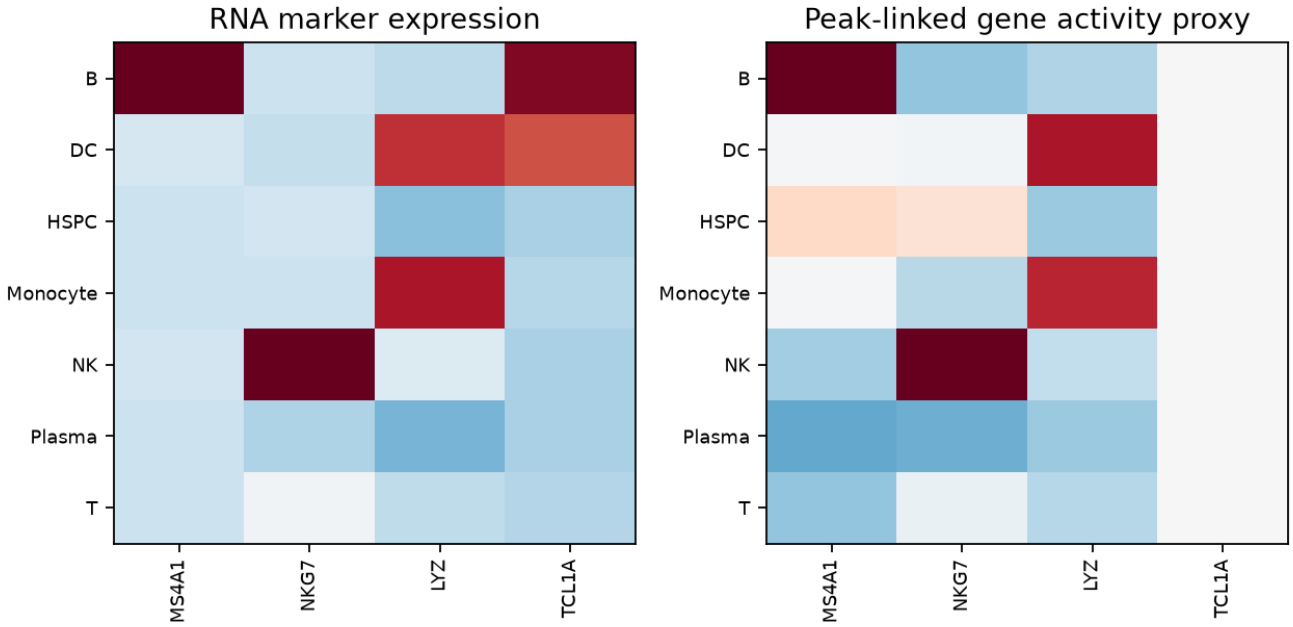
Canonical R WNN cross-check on training cells

Training cells	Top-15 graph overlap
7171	0.195

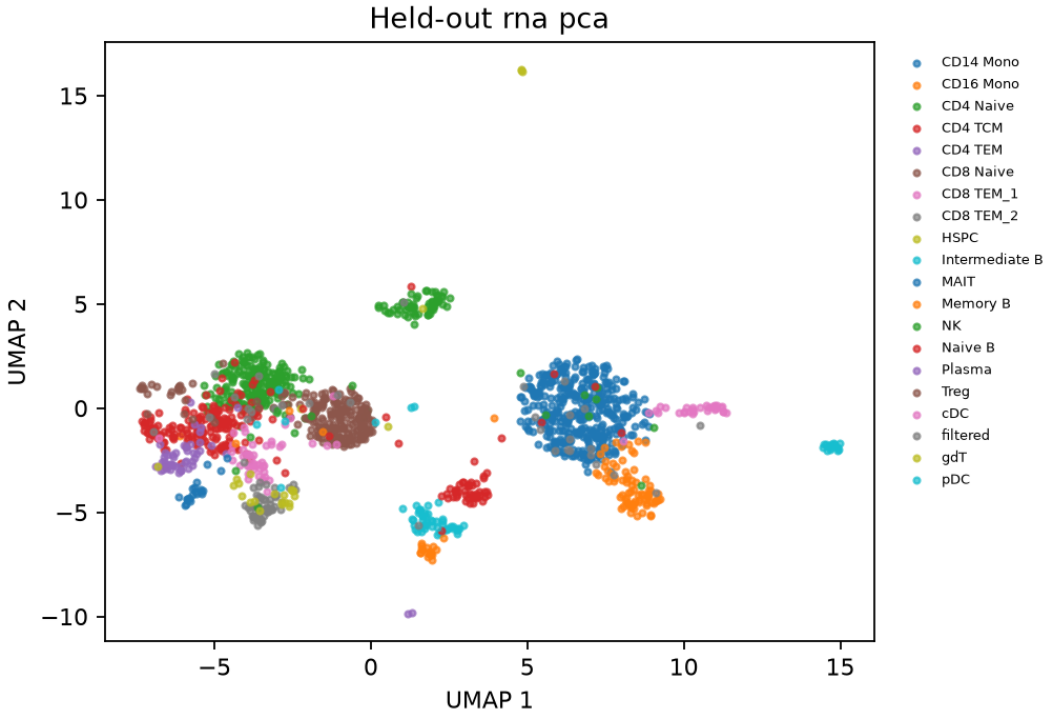
Alignment is descriptive where per-modality spaces are defined; baseline projections are reference spaces, not learned contrastive towers. Masking index 0 denotes the first configured modality and index 1 the second. Seed rows measure training randomness; neighbor overlap measures adapter refits on 80% training samples while retaining fixed train-only preprocessing. Nearest-neighbor label agreement uses labeled queries in the declared pool; candidates without a reference label count as nonmatches. Luminal diagnostics, when present, restrict true labels to LumA/LumB and retain the frozen four-class predictions, including errors into other classes. They are descriptive subgroup comparisons without separate model selection or subgroup confidence intervals.



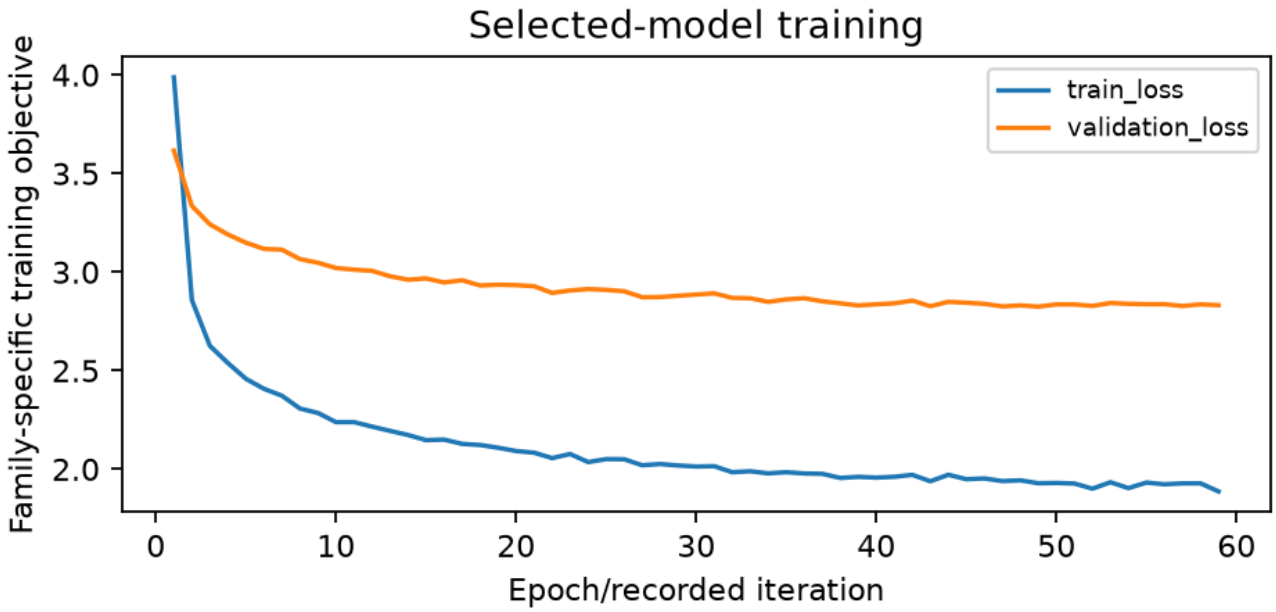
atac lsi umap. UMAP is fitted on training samples and transforms held-out samples; visualization only.



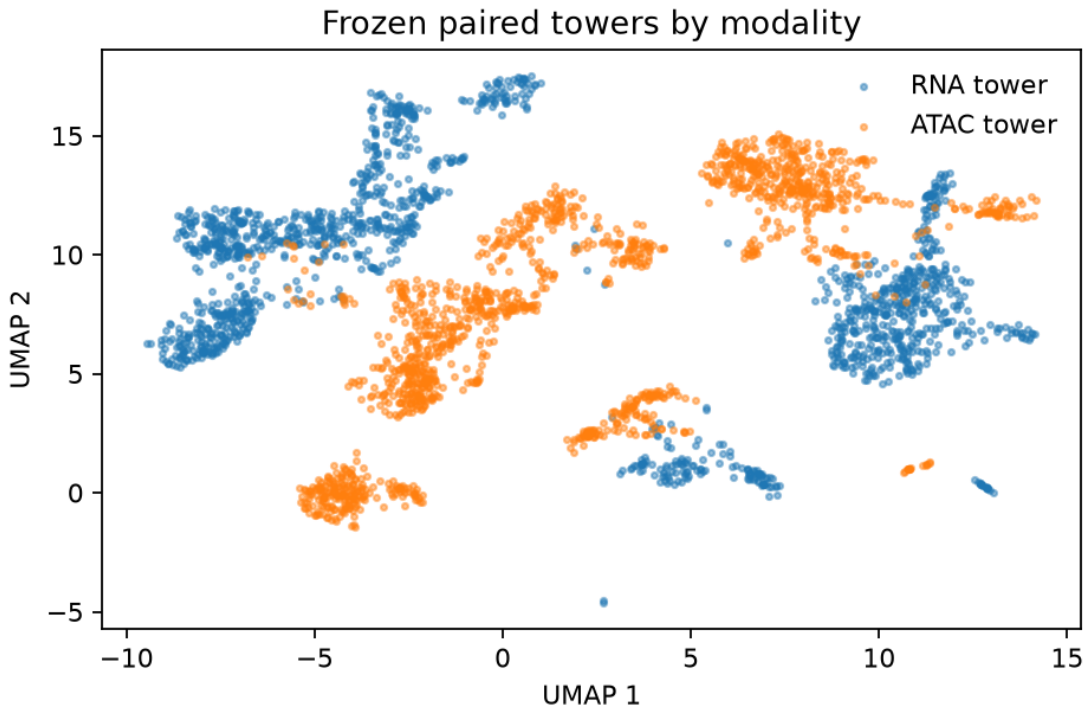
marker gene activity. Each marker is standardized across the displayed coarse cell groups within each panel: red is higher and blue is lower relative signal. Peak-linked activity is a vendor-annotation proxy, not measured expression or independent label validation.



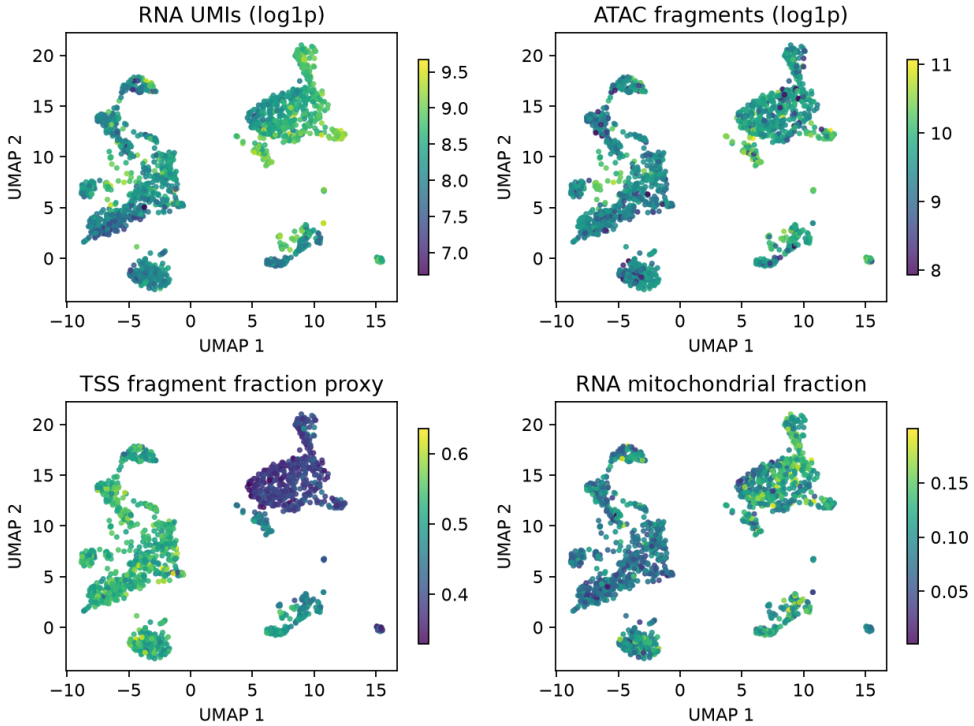
rna pca umap. UMAP is fitted on training samples and transforms held-out samples; visualization only.



selected training curves. Training and validation objectives from the selected frozen run; selection used validation only.



modality umap. Both frozen tower embeddings share one display map, fitted on training RNA and ATAC embeddings and used to transform held-out paired cells. Modality mixing is descriptive; quantitative retrieval and FOSCTTM supply the alignment evidence.



selected qc umap. The same joint UMAP coordinates are colored by the recorded QC measurements. Each panel has its own scale; these displays are descriptive and do not separate biological variation from technical effects.

Permitted interpretation, limitations and reproducibility

- Single donor: no donor generalization
- Curated FixFlip labels derive from WNN clusters; WNN structural advantage; evaluation-only labels
- Single-donor cell-level results do not support donor generalization.
- Agreement with WNN-derived labels is not independent evidence of biological superiority to WNN.

Representations include sample IDs, split membership, model/checkpoint/preprocessing hashes and embedding definitions. The GPU ledger, environment locks, run manifests, decision log, split manifest and evidence source map accompany the outputs.

Sources: local data/label/model provenance manifests; pre-registered METHOD_NOTES.md; scvi-tools MultiVI, muon WNN, MOFA+, paired contrastive objectives and gated MIL implementations documented in the source and environment locks. Full citations are in the project report and docs/REFERENCES.json.